

Observer Patch Holography — An Independent Investigation

A verification-and-extension study of the OPH framework (FloatingPragma/observer-patch-holography). Dated 2026-07-13. Methods: independent symbolic recomputation, controlled source-code experiments, and machine-checked proofs — no reliance on OPH's own scripts for any headline claim.

Executive summary

Observer Patch Holography (OPH), by Bernhard Mueller, is a proposed "observer-consistency theory of everything": no observer sees the whole world; each holds a local patch; **physics is the public fixed point that survives agreement across overlapping patches**. It claims to recover General Relativity and the Standard Model from a handful of axioms plus one number.

After independently checking its headline numbers, auditing the source code behind them, examining the formal proofs, and stress-testing the boldest applications, the verdict is:

OPH is neither crank numerology nor the "true theory of everything" it advertises. It is a serious framework with one genuinely impressive physics result (the cosmological-constant magnitude), a logically sound consensus core (which this study completes and extends past the authors' own proofs), and a presentation layer that systematically oversells conditional results as parameter-free — with the applications layer containing an outright internal contradiction.

Crucially, the honesty is not uniform. There is a clear **gradient**: OPH's technical corpus (papers, claim registry, theorem gates, negative-control lists) is scrupulous about what is *not* derived; the README slightly oversells; and the popular/applications layer is the boldest and least evidenced.

This study also makes **constructive contributions**: five machine-checked proofs that complete and generalize OPH's sorry-bearing formal core across three layers (algebraic, multi-observer, dynamical), plus a new sharp falsifier the framework does not state.

Method

Every headline claim was re-derived from authoritative inputs (PDG/NIST/Planck) rather than trusted from OPH's outputs. Three independent instruments were used:

- `symbolicaql` — a symbolic-math engine, for recomputing OPH's algebraic identities and closed forms from scratch.
- **OPH's own solver, run as a controlled experiment** — the α /electroweak chain (`code/P_derivation/`, `pure-Decimal` stdlib) was executed with single constants swapped, isolating one assumption at a time.
- `spleanql` — a dependent-type-theory proof kernel (no tactics, explicit recursors), for machine-checking the formal claims and building extensions.

Findings were tracked as hypotheses with confidence "mass" in a `labbookql` notebook (domain `oph`), each with a concluded experiment and committed evidence. The full ledger appears at the end.

Part I — What OPH is

OPH reconstructs physics from finite observer patches whose descriptions must agree where they overlap. Its four operational pillars: **Boundary** (each observer has limited local access), **Readback** (records become objective when comparable across boundaries), **Repair** (conflicting records generate pressure to update), **Continuation** (records determine where observers can stably exist).

Quantitatively, everything hangs on two fixed points:

- a local **pixel fixed point** $P^* \approx 1.6310$, solving $P = \phi + \sqrt{\pi} / A_T(P)$ (ϕ = golden ratio), which feeds α , gauge structure, and particle rows;
- a global **record-capacity fixed point** $N_{CRC} \approx 10^{12.2}$, feeding cosmology.

The repository is large and actively developed ($\approx 1,190$ commits, 878 Python files, 19 Lean files, six papers, a book, a claim registry, runnable verification code).

Part II — Independent numerical verification

Recomputed from authoritative inputs on `symbolicaql`:

Quantity	OPH formula	Independent result	vs. experiment	Verdict
Spectral tilt	$n_s = 1 - P^*/48$	0.9660215	0.27σ from Planck	✓ clean
Cosmological constant	$N_{CRC} = \pi \cdot \exp(6\pi/(P \cdot \alpha_U))$	$\log_{10} = 122.55$	7% from observed dS entropy	✓ striking
MOND scale	$a_0 = (15/8\pi^2)c^2\sqrt{(\Lambda/3)}$	$1.03 \times 10^{-10} \text{ m/s}^2$	ratio 0.86 to empirical	✓ within 15%
Proton spin	$C_F/(C_F + C_A) = 4/13$	0.3077	0.59 σ from 0.29 $\pm$ 0.03	✓
Koide phase	$\delta = 2/9$	0.22222	see below	~ oversold precision
Baryon ratio	$(1/6)^{12}$	4.59×10^{-10}	factor 1.33 low	~ order-of-magnitude
Fine structure α^{-1}	$\sqrt{\pi}/(P - \phi)$	136.9948 (source)	needs +0.041 hadron closure	✗ see Part III

Koide caveat (new finding). Recomputing the phase from raw PDG lepton masses gives $\delta = 0.222221$ (electron anchor) to 0.222270 (τ anchor), because Koide's $Q = 0.6666605$ is not *exactly* $2/3$. So $\delta \approx 2/9$ holds to ~ 4 significant figures, but the advertised "match to 4 ppm ($0.2222248 \pm 6.3e-6$)" is convention-dependent and overstates the robustness.

Part III — The α crux: a SUSY-conditional knife-edge

OPH's flagship claim is deriving α ("from one number"). Reading the actual solver (`code/P_derivation/paper_math.py`) shows the fixed point is **genuine** — $A_T(P)$ is built from P via group theory and RG running with **no measured coupling, mass, or angle used as a target**. So α is *not* a naive fit. But two things the marketing omits are load-bearing:

1. It structurally assumes supersymmetry. The running uses $b_{mssm} = (33/5, 1, -3)$ — the *MSSM* beta coefficients. A controlled experiment (OPH's own solver, only the betas swapped to the Standard Model values ($41/10, -19/6, -7$)):

Betas	α^{-1}	closure
MSSM (OPH)	136.99483 (reproduces documented value)	✓

Betas	α^{-1}	closure
SM (physical)	—	fails: no valid m_Z fixed point

The SM's negative b_2, b_3 drive the inverse couplings through zero inside the solver's running range ($\alpha_2^{-1} = -0.20$), so α goes negative and the construction produces *no α at all*. With the physically-correct Standard Model spectrum, the result does not merely change — it ceases to exist.

2. $\alpha = 137$ is a knife-edge in particle-content space. Mapping α^{-1} over the beta coefficients (a scan OPH never publishes):

b_2 ($b_3 = -3$)	2.0	1.5	1.0 (MSSM)	0.7	0.4	0.0
α^{-1}	123.3	129.7	137.0	141.9	147.2	155.0

Sensitivity $d\alpha^{-1}/db_2 \approx -16$, $d\alpha^{-1}/db_3 \approx -12$ per unit; the measured value is hit *only* at the MSSM integers, and the construction fails outside a narrow band. **$\alpha = 137$ is inherited from the assumed MSSM spectrum on a steep slope — the α prediction is exactly as strong as OPH's (unproven) claim to derive supersymmetry, no stronger.**

3. The "exact" α is empirically closed. The strict source-only value is 136.9948 ($\sim 0.03\%$ off); reaching 137.036 requires adding empirical $e^+e^- \rightarrow \text{hadrons}$ data. OPH's own SOURCE_SPECTRAL_THEOREM.md concedes the needed hadronic spectral measure is "absent from the current corpus."

A new falsifier (constructive). Because $\alpha = 137 \Leftrightarrow$ MSSM betas, OPH gains a sharp test it does not state: *if collider data establishes SM-like (non-MSSM) running of the couplings, OPH's α prediction is falsified*. This turns the SUSY-dependence critique into a Popperian virtue.

Electroweak masses. The same chain's "0.02–0.08% agreement" for $W/Z/\text{Higgs}$ masks that, at real measurement precision, Z is -34.5σ , the strict Higgs $+5.8\sigma$, and W -2.95σ from data. Only the top ($+0.10\sigma$) and Λ_{QCD} (-0.27σ) are genuine precision matches.

Part IV — The cosmological constant: OPH's strongest result

OPH's capacity fixed point is $N_{\text{CRC}} = \pi \cdot \exp(6\pi / (P \cdot \alpha_U))$, built from the same source quantities as α . Independently:

- OPH: **$\log_{10} N_{\text{CRC}} = 122.55$** ($\sim 3.5 \times 10^{122}$)
- Observed de Sitter entropy $S = 3\pi / (\Lambda_{\text{P}}^2)$: **$\log_{10} S = 122.52$** ($\sim 3.3 \times 10^{122}$)

They agree to **7%**. OPH reproduces the *magnitude* of the cosmological constant — the notorious 120-orders-of-magnitude problem — from the α fixed point. That is more than most theories achieve.

The result is hyper-sensitive to the integer "6" (≈ 20 orders of magnitude per unit; the value needed to hit Λ exactly is 5.9986, and OPH uses 6). Auditing that integer's provenance largely vindicates it: **$6 = m_{\text{rep}}/\beta_{\text{EW}} = 24/4$** , where **$24 = 2 \cdot \dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2 \cdot (8+3+1)$** (twice the SM gauge-algebra dimension) and **$4 = n_c + 1$** . OPH documents seven *rejected* alternative integers (including "24-from-SU(5)," rejected despite giving the same value) and a no-empirical-input ledger. So the integer is structurally derived, not tuned to Λ . Residual caveats: the $\times 2$ orientation-doubling axiom is the softest link, and the result inherits α_U 's MSSM-conditionality.

Part V — Formal-proof status, and a completed replacement

What OPH actually proves. Its Lean (Lean/ObserverPatchHolography/) does *not* machine-check its central Proposition 4.2. The repair operator itself is literally sorry:

- `localRepair` (Primitives.lean:129), `Repair` (:133), and `repair_respects_gauge` are sorry.
- What *is* sorry-free: generic Newman's-lemma abstract rewriting over an *opaque* relation (self-labeled "a skeleton, not a Prop 4.2 statement"), boundary reconstruction over one concrete demo carrier, and a two-bit example. The authors state this explicitly (Primitives.lean:1027–1032).

So the "Lean-verified consensus core" reduces to standard textbook rewriting plus a demo, with the dynamical heart deliberately unproven.

What this study proves instead. Five `splean` artifacts (all axiom-free modulo the carrier) complete and generalize the core across three layers:

Layer	File	Content
Toy	<code>oph_consensus_core.splean</code>	2-state consensus: convergence, idempotence, overlap agreement
Concrete fill	<code>oph_repair_prop42.splean</code>	a concrete (non-sorry) repair on a two-patch net discharges consistency, faithfulness, idempotence, and the gauge congruence OPH leaves sorry
Algebraic (general)	<code>oph_repair_general.splean</code>	Prop 4.2 for any carrier from one read-after-write law $r(w \vee v)=v$; concrete witness proves non-vacuity
Multi-observer	<code>oph_consensus_extended.splean</code>	overlap consensus, universal agreement on public states, transitive propagation $\rightarrow$ the public world is globally well-defined
Dynamical	<code>oph_repair_dynamical.splean</code>	bounded Lyapunov convergence: from any state, $\Phi(s)$ repair steps reach consensus ($\Phi=0$), by induction on the potential

Two structural insights fall out of the `#axioms` receipts, neither stated by OPH:

- **Two-layer decomposition.** Gauge-invariance and consensus (`repair_respects_gauge`, `overlap_consensus`, `consensus_transitive`) need *no* coherence law — they are pure congruence. Only convergence/faithfulness use read-after-write. This mirrors physics (gauge symmetry exact; equilibration conditional) and suggests a cleaner OPH axiomatization.
- The dynamical convergence — the piece OPH leaves sorry and earlier work modeled only as a one-shot retraction — is genuinely provable from a single descent law.

Together these constitute a machine-checked OPH consensus/repair theory that **strictly exceeds OPH's own (sorry-bearing, single-carrier) Lean**.

Part VI — The applications layer (OMEGA)

The website (omega.floatingpragma.io) and `APPLICATIONS.md` claim tabletop fusion, anti-gravity hoverbikes, AGI, and compute. All are honestly labeled *Draft / pre-hardware / "not measured evidence."* Two observations:

- **Internal contradiction.** APPLICATIONS.md:58 claims a ~\$1,000 build "would likely mine a Bitcoin block in milliseconds." That requires candidate enrichment of $\sim 10^{13}$ even at an absurdly optimistic 10^{12} optical evals/s — i.e., *solving* the proof-of-work search. But the README insists the compute lift is modest and "the classical complexity-class problem remains untouched." The boldest OMEGA claim contradicts OPH's own stated limitation (and SHA-256d has no exploitable pre-evaluation structure to enable it).
- **No hardware evidence.** The one place OPH ran real hardware, the IBM Quantum Cloud archive (frozen 2026-07-11), states plainly: "No experiment in this bundle distinguishes OPH from standard quantum mechanics."

Part VII — The meta-pattern

The single most useful finding for a reader deciding how to weigh OPH:

The honesty runs corpus → README → applications. The technical papers are rigorous and self-critical — they explicitly disclaim neutrinos (a "rejected target-informed candidate" from a "hand-written template" that NuFIT rejects), baryogenesis/sphaleron ("no baryogenesis theorem package"), quark masses (proven non-identifiability), and the Yang-Mills gap (conditional). The misleading impression comes from popular headlines and isolated close-looking numbers, not the documents. The two places even the README oversteps are the α "zero-input" framing and the "matches the W/Z/Higgs" phrasing.

The right critique target is the marketing, not the corpus.

Part VIII — Simulation, and the accommodation pattern

The analytical study was followed by **four independent simulations** and a native **3D app**, so the mechanism could be *run*, not just read.

- **Consensus dynamics** (oph_consensus_sim.py): OPH's repair reaches the public fixed point on every topology, and the convergence **rate = the patch-net's spectral gap** (algebraic connectivity) — classical Laplacian-consensus theory. This *measures* the dynamical-rate law OPH leaves abstract (the "form"), while the cosmological *value* still needs the physical screen graph.
- **Dark matter, 2D** (oph_darkmatter_sim.py, ..._interaction.py): a topological defect leaves an **unrepairable residual** only where repair fails, with an extended $\sim 1/r^2$ (isothermal-halo) profile and a logarithmic (deep-MOND) two-defect interaction. Qualitatively supportive of OPH's "dark stress = imperfect repair."
- **Dark matter, genuine 3D** (oph-sim-macos, an interactive SwiftUI/SceneKit app): the decisive dimensional-lift test. A vortex **line** (2D-symmetric) reproduces the isothermal halo ($p \approx -2.4$); a **localized vortex loop** gives $p \approx -6.5$ — a compact, Newtonian (dipole) knot, *not* a dark-matter halo. **The MOND-shaped halo was a 2D/line-symmetry artifact; it does not survive to a localized 3D defect.** This *revised the earlier supportive finding downward* — the discipline held even against OPH.

How OPH explains a result it did not predict

That 3D result is the sharpest single example of the pattern the whole study found. *Can OPH explain it?* Yes — and grounded in its own text: OPH's dark stress is a **"collar remainder"** built on **extended, codimension-1 structures** (collars, quotient-edges, finite-thickness bands — 84 "collar" mentions in the dark-matter paper, with a thickness hierarchy $\square_{UV} \square \delta_{\text{collar}} \square \square$). That

is the extended defect the 3D sim shows produces an isothermal halo; the localized loop that came out Newtonian is simply not OPH's object.

But the explanation is **accommodation, not prediction**:

- 1 OPH *assumed* the extended collar; it never stated that *localized* defects fail. The simulation supplies the missing "why."
- 2 OPH is holographic — everything lives on 2D screens — so codimension-1 is *baked in*. Consistency with an extended-defect result is partly structural inevitability, not a risky bet. **The machinery is rich enough to accommodate the outcome, which is exactly why OPH is hard to falsify.**

The constructive flip: this sharpens into one genuine, falsifiable claim OPH should adopt explicitly — *the dark stress is codimension-1, so dark matter traces extended, web-like structure (filaments and sheets), not isolated point clumps* — which loosely matches the observed cosmic web. A negative simulation result thus becomes the flip side of a real OPH-flavored prediction.

This is the framework in miniature: OPH explains, by a construction flexible enough that "explains" means "was built to accommodate" more than "forecast" — while still yielding one crisp, checkable commitment.

Evidence ledger

Eighteen hypotheses (labbookql, domain oph), each with a concluded experiment and committed evidence. Confidence "mass" reflects the weight of evidence after this study.

Mass	Hypothesis
0.90	α/EW chain uses no measured target but is conditional on assumed MSSM/SUSY betas
0.90	Prop 4.2 sorry-gap is fillable (concrete repair, gauge congruence proved)
0.85	Dynamical Prop 4.2 (iterated repair $\rightarrow$ consensus) machine-provable via Lyapunov descent
0.80	$\alpha^{-1} \approx 137$ is a knife-edge selecting the MSSM integers
0.80	Multi-observer thesis machine-checkable, extending the single-carrier core
0.70	Spectral tilt $n_s = 1 - P^*/48$ matches Planck (0.27σ)
0.70	Consensus/repair core is a consistent machine-checkable structure
0.70	Cosmological-constant magnitude ($\sim 10^{122}$) is a genuine structural prediction
0.65	MOND scale a_0 matches empirical (within 15%)
0.60	Proton-spin fraction $4/13$ matches (0.59σ)
0.55	Koide phase $\delta = 2/9$ (real, but ppm precision is convention-dependent)
0.45	Baryon ratio $(1/6)^{12}$ (order-of-magnitude only, factor 1.33 low)
0.40	α^{-1} is a zero-input prediction (genuine fixed point, but MSSM-conditional + hadron-incomplete)
0.25	Strict W/Z/H masses match at measurement precision (no — tens of σ)
0.25	Central consensus dynamics is machine-checked in OPH's Lean (no — it's sorry)

Mass	Hypothesis
0.25	OPH predicts neutrino masses/mixing from source (no — rejected template)
0.20	OPH derives the sphaleron/baryogenesis value (no — disavowed in the corpus)
0.20	Bitcoin-in-ms compute claim is consistent with OPH's complexity limitation (no — contradiction)

Limitations of this study

- The dynamical convergence theorem is proved for an *abstract* potential and descent law; it does not construct a concrete non-trivial repair operator over general finite patch-nets with an *async schedule* and a derived potential — that fuller mechanization remains open (as it is in OPH).
- The α controlled experiment used reduced group-theory cutoffs (applied equally to both branches); it reproduces the documented MSSM value to 5+ digits, so the MSSM-vs-SM contrast is faithful, but the absolute reduced-cutoff numbers are not the full-precision values.
- The cosmological-constant provenance audit vindicates the integer 6 up to the orientation-doubling axiom, which was not independently re-derived here.
- No independent physical experiment was performed; "verification" here means recomputation, code audit, and proof, not laboratory test.

Reproducibility

- **OPH repo:** FloatingPragma/observer-patch-holography (cloned into this working tree).
- **Proof artifacts (this study):** verification/*.splean (five files), checked with spleanql <file>.
- **Controlled experiments:** sm_vs_mssm.py, beta_scan.py (patch paper_math.PaperMathContext.b_mssm; pure stdlib, no venv needed).
- **Symbolic checks:** symbolicaql eval (fine structure, Koide, N_CRC, MOND, σ -distances).
- **Logbook:** labbookql query "SELECT mass,label FROM hypotheses WHERE domain='oph' ORDER BY mass DESC".

Conclusion

OPH deserves to be taken more seriously than a typical "theory of everything" and less literally than its marketing. Its consensus/repair core is a coherent, now fully machine-checked idea; its cosmological-constant result is genuinely impressive and survives provenance scrutiny; and its papers are honest about their many open gaps. But its flagship α claim is a supersymmetry-conditional knife-edge dressed as parameter-free, its electroweak "agreements" are many-sigma at real precision, and its applications layer overreaches to the point of self-contradiction.

The most durable output of this study is not the verdict but the **method and the extensions**: independent recomputation over trust, controlled single-variable experiments on the authors' own code, and a completed formal core that carries OPH's central thesis further than its authors did — including a concrete, testable falsifier they can adopt.